

Individual variation and intraclass correlation in arachidonic acid and eicosapentaenoic acid in chicken muscle

Chicken meat with reduced concentration of arachidonic acid (AA) and reduced ratio between omega-6 and omega-3 fatty acids has potential health benefits because a reduction in AA intake dampens prostanoid signaling, and the proportion between omega-6 and omega-3 fatty acids is too high in our diet. Analyses for fatty acid determination are expensive, and finding the optimal number of analyses to give reliable results is a challenge. The objective of the present study was i) to analyse the intraclass correlation of different fatty acids in five meat samples, of one gram each, within the same chicken thigh, and ii) to study individual variations in the concentrations of a range of fatty acids and the ratio between omega-6 and omega-3 fatty acid concentrations among fifteen chickens. Fifteen newly hatched broilers were fed a wheat-based diet containing 4% rapeseed oil and 1% linseed oil for three weeks. Five muscle samples from the mid location of the thigh of each chicken were analysed for fatty acid composition. The intraclass correlation (sample correlation within the same animal) was 0.85-0.98 for the ratios of total omega-6 to total omega-3 fatty acids and of AA to eicosapentaenoic acid (EPA). This indicates that when studying these fatty acid ratios, one sample of one gram per animal is sufficient. However, due to the high individual variation between chicken for these ratios, a relatively high number of animals (minimum 15) are required to obtain a sufficiently high power to reveal significant effects of experimental factors (e.g. feeding regimes). The present experiment resulted in meat with a favorable concentration ratio between omega-6 and omega-3 fatty acids. The AA concentration varied from 1.5 to 2.8 g/100 g total fatty acids in thigh muscle in the fifteen broilers, and the ratio between AA and EPA concentrations ranged from 2.3 to 3.9. These differences among the birds may be due to genetic variance that can be exploited by breeding for lower AA concentration and/or a more favorable AA/EPA ratio to produce meat with health benefits.